Practical 8

Aim: To implement **FOSS cloud functionality using VSI** by performing:

a) Creating a **Virtual Machine (VM)** using an available OS
b) Creating a **Storage Pool by allocating memory**

Software Required

- Linux (Ubuntu recommended)
- **KVM**
- **Virtual Machine Manager (virt-manager)**
- Internet connection

(KVM should already be installed from Practical 7)

Step 1 Start Virtual Machine Manager

Open **Terminal** and run: virt-manager

The **Virtual Machine Manager GUI** will open.

Step 2 Create a Virtual Machine

Click: Create New Virtual Machine

Select: Local install media (ISO Image)

Click **Forward**

Step 3 Select Operating System

Choose OS ISO file.

Example: ubuntu-22.04.iso

Click **Forward**

Step 4 Allocate Memory and CPU

Example configuration:

RAM : 2048 MB

CPU : 2

Click **Forward**

Step 5 Create Virtual Disk

Create disk storage:

Storage Size : 20 GB

Click **Forward**

Step 6 Name the Virtual Machine

Example:

VM Name : CloudVM

Click **Finish**

The VM will start installing the OS.

Step 7 Open Storage Management

In **Virtual Machine Manager**

Go to: Edit → Connection Details

Open the **Storage Tab**

Step 8 Create Storage Pool

Click: Add Pool

Fill details:

Name : CloudStorage

Type : Directory

Target Path : /var/lib/libvirt/images

Click **Finish**

Storage pool created ✅

Step 9 Create Storage Volume

Inside the pool click: Add Volume

Fill:

Volume Name : cloud_disk

Capacity : 10 GB

Click **Finish**

Storage volume created.

Result

- Virtual Machine created successfully
- Storage pool created
- Storage volume allocated for VM

This demonstrates **basic FOSS cloud infrastructure using VSI**.